

Jaral Arroyo-Jefferson

(786) 812-1907 | jaral.arroyo.jefferson@gmail.com | [linkedin.com/in/jaral-arroyo-jefferson/](https://www.linkedin.com/in/jaral-arroyo-jefferson/) | jaralaj.me | github.com/JaralAJ

EDUCATION

Case Western Reserve University

Bachelor of Science in Aerospace and Mechanical Engineering, Minor in Computer Science

Cleveland, OH

May 2029

EXPERIENCE

Undergraduate Robotics Research Assistant

Oct. 2025 – Present

Cybernetics and Physical Intelligence Laboratory — Human Fusions Institute

Cleveland, OH

- Designed Python control interface using Dynamixel SDK for a cable-driven robotic tentacle arm with 5 preset configs, cutting trial setup time by 2 min and adopted by 3 lab researchers
- Built real-time digital twin in MuJoCo to validate position/orientation estimation algorithm against physical hyper-redundant cable-driven arm
- Implemented 100Hz Extended Kalman Filter fusing motor torque, cable length, and Vicon mocap data for real-time orientation estimation

Optical Tweezers Research Project Engineering Lead

Aug. 2023 – May 2025

Ransom Everglades School

Miami, FL

- Led team of 6 to modify OpenFlexure 3D inverted microscope design, enabling laser integration for low-cost optical tweezing system
- Designed parts in Fusion 360 for 3D printing, assembled microscope, and integrated electronics using Arduino and Raspberry Pi
- Presented research on optical tweezers and microscopy at the 2025 APS Global Physics Summit

NASA Aerial Robotics Research Intern

May 2024 – July 2024

NASA STEM Enhancement in Earth Science Internship

Austin, TX

- Fused IMU and LiDAR sensor data using a Kalman Filter, reducing IMU/GPS noise by 24% and LiDAR estimation error by 50%
- Assembled drone hardware and configured flight software and sensors, serving as flight test pilot for data collection
- Processed binary sensor data using Python and pyulog, presented findings at the 2024 American Geophysical Union Annual Meeting

PROJECTS

ML Star Classification | *Python, XGBoost, Tableau, Colab*

Dec. 2025

- Built XGBoost model to predict star type from stellar attributes, including temperature, luminosity, and magnitude
- Created Tableau and Colab visualizations to identify defining characteristics across stellar evolutionary stages

NASA Meteorite & Bolid: Data Analysis Report | *Python, Pandas, Matplotlib, Tableau*

Jan. 2024

- Cleaned and merged two NASA Open Data Portal datasets in Python using Pandas, imputing missing values and standardizing coordinate formats
- Conducted exploratory data analysis using Matplotlib and Tableau, uncovering a strong linear relationship between radiated and impact energy

LEADERSHIP

SHPE Vice President of Communications

Sept. 2025 – Present

Society of Hispanic Professional Engineers (SHPE), CWRU Chapter

Cleveland, OH

- Manage chapter communications and social media for a 5-member executive board
- Led outreach and promotion for Dia de Ciencias, engaging 25 local high school students

TECHNICAL SKILLS

Programming & Robotics: Python, MATLAB, Git, GitHub, HTML, CSS

Controls & Estimation: Kalman Filtering, Extended Kalman Filter, Sensor Fusion (IMU, LiDAR, Motion Capture), State Estimation, MuJoCo

Engineering & Design: SolidWorks, Fusion 360, 3D Printing, Laser Cutting, CNC Machining, Arduino, Raspberry Pi

Data Science: Machine Learning (XGBoost, Random Forest, Logistic Regression), Pandas, Matplotlib, Tableau

Languages: Bilingual Spanish